WORKSHEET

Chapter 2: Multiplying and dividing

1 Simplify fully:

a $3 \times 4x$
b $2k \times 8$
c $-7 \times 7a$
d $-9b \times 2$
e $8x \times (-7)$
f $12ab \times 2$
g $7 \times 6x^2$
h $(-6) \times 2r^2$
i $-6x \times (-3)$
j $-4 \times 3x \times (-5)$

2 Simplify fully:

a $2 \times 3b$
b $4m \times n$
c $7x \times x$
d $-6b \times 8d$
e $5a \times (-3b)$
f $-4m \times 8n$
g $-7b \times (-5e)$
h $8x \times 5x$
i $-5s \times 6s$
j $7x \times 3y \times 2x$

3 Simplify fully:

a $2b \div 2$
b $10m \div 5$
c $8k \div 4$
d $9x^2 \div 3$
e $-15y \div 5$
f $20a \div (-4)$
g $(-27r) \div (-9)$
h $100m^2 \div 20$
i $(-50w^2) \div 25$
j $-49p \div (-7)$

4 Express as a fraction and then simplify fully:

a $2x \div 8$
b $5a \div 15$
c $20e \div 5e$
d $8b \div 32b$
e $x^2 \div x$
f $x \div x^2$
g $25x^2 \div 15x$
h $8xy \div 10yz$
i $16a^2 \div 12a^2\text{b}$
j $24m^2n \div 18mnp$

5 Simplify fully:

a $3xy^2 \times x$
b $2mn \times np$
c $-5a^2b \times abc$
d $7pq \times 3p^2$
e $4st \times (-5st)$
f $2kj \times 8kj^2$
g $-8xyz^2 \times (-5xz)$
h $12abc \times 4cde$
i $-20v^2 \times 3vw^2$
j $9m^2p \times (-8m^2np^2)$

6 Simplify fully:

a $\frac{6x}{3}$
b $\frac{5m}{15}$
c $\frac{8}{2k}$
d $\frac{20v}{35w}$
e $\frac{18bc}{12c}$
f $\frac{15x}{45x^2}$
g $\frac{16m}{24n^2}$
h $\frac{12pr}{8pq}$
i $\frac{9xyz}{15x}$
j $\frac{27acd}{18bcd}$
k $\frac{28x^2}{35xy}$
l $\frac{17st^2}{34rst}$
m $\frac{12v^2w}{20wx}$
n $\frac{27ab^2}{24abc}$
o $\frac{40x^3}{48x}$

7 Simplify fully:

a $\frac{4x}{5} \times \frac{1}{3}$
b $\frac{2a}{3b} \times \frac{7}{c}$
c $\frac{5m}{3} \times \frac{3}{8}$
d $\frac{2a}{5} \times \frac{4}{3a}$
e $\frac{7w}{6} \times \frac{3}{w}$
f $\frac{2k}{5} \times \frac{3k}{4}$
g $\frac{10}{3b} \times \frac{9}{15}$
h $\frac{7}{2a} \times \frac{8a}{11}$
i $\frac{10m}{8} \times \frac{4n}{15m}$
j $\frac{16x}{yz} \times \frac{7z}{12x}$

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Answers

1 **a** $12x$
b $16k$
c $-49a$
d $-18b$
e $-56x$
f $24ab$
g $42x^2$
h $-12r^2$
i $18x$
j $60x$

2 **a** $6b$
b $4mn$
c $7x^2$
d $-48bd$
e $-15ab$
f $-32mn$
g $35be$
h $40x^2$
i $-30s^2$
j $42x^2y$

3 **a** b
b $2m$
c $2k$
d $3x^2$
e $-3y$
f $-5a$
g $3r$
h $5m^2$
i $-2w^2$
j $7p$

4 **a** $\frac{x}{4}$
b $\frac{a}{3}$
c 4
d $\frac{1}{4}$
e x
f $\frac{1}{x}$
g $\frac{5x}{3}$
h $\frac{4x}{5z}$
i $\frac{4}{3b}$
j $\frac{4m}{3p}$

5 **a** $3x^2y^2$
b $2mn^2p$
c $-5a^3b^2c$
d $21p^3q$
e $-20s^2t^2$
f $16k^2j^3$
g $40x^2yz^3$
h $48abc^2de$
i $-60v^3w^2$
j $-72m^4np^3$

6 **a** $2x$
b $\frac{m}{3}$
c $\frac{4}{k}$
d $\frac{4v}{7w}$
e $\frac{3b}{2}$
f $\frac{1}{3x}$
g $\frac{2m}{3n^2}$
h $\frac{3r}{2q}$
i $\frac{3yz}{5}$
j $\frac{3a}{2b}$
k $\frac{4x}{5y}$
l $\frac{t}{2r}$
m $\frac{3v^2}{5x}$
n $\frac{9b}{8c}$
o $\frac{5x^2}{6}$

7 **a** $\frac{4x}{15}$
b $\frac{14a}{3bc}$
c $\frac{5m}{8}$
d $\frac{8}{15}$
e $3\frac{1}{2}$
f $\frac{3k^2}{10}$
g $\frac{2}{b}$
h $2\frac{6}{11}$
i $\frac{n}{3}$
j $\frac{28}{3y}$

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